# PATHUM LAKSHAN

FULL-STACK SOFTWARE ENGINEER

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## PROFESSIONAL SUMMARY

Full-stack engineer with 2+ years building production systems for retail, transport, banking and manufacturing clients. Backend depth in Django and Laravel, frontend work in React and Vue, and end-to-end ownership from database design through Linux deployment and ongoing maintenance. Currently operating a live multi-module ERP as a company's system of record, alongside final-year research in multimodal deep learning for financial time series. Experienced working remotely with distributed teams across European hours.

## PROFESSIONAL EXPERIENCE

### **Independent Software Consultant** | Remote - Sri Lanka

*January 2026 – Present*

Design, build and operate production business systems for SME clients, owning the full lifecycle from requirements gathering and schema design through deployment, monitoring and ongoing maintenance on self-managed Linux infrastructure.

**Senovka ERP - Senovka Plastics** - *Django, Gunicorn, Nginx, Chart.js, SQLite*

- Designed and built a multi-module ERP covering inventory, procurement, petty cash and vehicle logging, replacing spreadsheet-based processes across 2 departments and now serving as the company's live system of record.
- Deployed and maintain the platform on a self-managed Ubuntu VPS: Nginx reverse proxy, Gunicorn over Unix socket under systemd, Let's Encrypt SSL, and a scripted Git-based deploy pipeline reducing releases to a single command.
- Built role-based access control and interactive dashboards giving management daily visibility into cash position and stock movement.
- Supports 3 daily users across 16 modules; live in production at senovkaplastics.cloud.

**Procurement Portal - Senovka Plastics** - *Django*

- Separate purchase-order and supplier-management application deployed on an isolated subdomain and service port, sharing the same VPS infrastructure under independent systemd services.

**Luxe Salon - Booking & Scheduling Platform** - *Flask, Gunicorn, Nginx*

- Multi-practitioner booking system with owner-managed scheduling, concurrency-safe slot locking to prevent double-booking, and automated end-of-day revenue reporting.

**Employee Attendance & Overtime Tracker** - *JavaScript, SheetJS*

- Browser-based tool parsing raw biometric device exports into payroll-ready overtime calculations - day-type rules, overnight shift detection and ambiguous date-format resolution - with styled Excel export.

### **Junior Full-Stack Developer** | Jaan Network (Pvt) Ltd - Colombo 03, Sri Lanka

*March 2025 – December 2025*

**Hybrid POS System with Offline Sync** - *Flask, MySQL*

- Built a point-of-sale system covering supplier management, batch tracking, sales reporting and returns, deployed across 12 retail branches.
- Implemented an offline-first architecture with local MySQL instances and automatic reconciliation on reconnect, keeping transactions running through connectivity loss and resolving cross-branch data conflicts on sync.

**Transport Booking & Management Platform** - *Vue.js, Laravel*

- Developed a multi-modal booking system for trains, buses and flights, integrating third-party carrier APIs for live schedule and fare data.
- Built client booking dashboards and a super-admin portal with role-based authentication covering 3 vendor accounts.

**Banking Platform** - *Angular, Laravel*

- Delivered account registration, KYC verification, fund transfer and live currency-rate features against a regulated banking workflow.
- Implemented user activity tracking and audit logging supporting fraud detection and compliance reporting.

### **Junior Full-Stack Developer** | Heaven's Code - Negombo, Sri Lanka

*August 2024 – February 2025*

- Built a real-time order tracking platform (Angular, Laravel) with live status updates, reducing inbound customer support queries.
- Developed a responsive restaurant website (HTML, Bootstrap, JavaScript) with interactive menus, galleries and an online reservation flow.

## RESEARCH

**Multimodal Cryptocurrency Direction Prediction** | BSc Final-Year Research, Saegis Campus
*2025 – 2026 | Supervisor: Mrs. Ashani Gallage*

Investigating whether fusing visual, temporal, retrieval and linguistic signals improve next-period directional prediction over single-modality baselines - an area where most published work relies on price series alone.

- **Visual -** CNN and YOLOv8-derived embeddings of candlestick chart segments, capturing formation patterns that numeric OHLCV features do not encode.
- **Temporal -** Hidden Markov Models identifying latent market regimes (trending, ranging, high-volatility) used as conditioning context for the classifier.
- **Retrieval -** FAISS vector index over historical embeddings, surfacing nearest-analogue market conditions as supporting evidence at inference time.
- **Language -** CryptoBERT and FinBERT sentiment extraction across crypto news and social streams, temporally aligned to the prediction window.
- **Fusion -** Combined multimodal representation feeding a directional classifier, benchmarked against price-only and single-modality baselines.

Stack: PyTorch, Hugging Face Transformers, FAISS, YOLOv8, hmmlearn, pandas, NumPy. Trained and evaluated on Apple Silicon.

## EDUCATION

**BSc (Hons) in Computer Science** | Saegis Campus, Sri Lanka
*2022 – 2026 (expected) | CGPA 3.72 / 4.00*

## TECHNICAL SKILLS

**Languages Python**, PHP, JavaScript, Java, C#, SQL

**Backend Django**, Django REST Framework, Flask, Laravel, Node.js / Express, Yii2

**Frontend React**, Vue.js, Inertia.js, Angular, React Native, HTML5, CSS3, Bootstrap

**Databases PostgreSQL**, MySQL, SQLite, MongoDB

**Infrastructure Ubuntu** server administration, Nginx, Gunicorn, systemd, Let's Encrypt / Certbot, VPS provisioning, Bash, Git-based deployment

**ML & Data PyTorch**, Hugging Face Transformers, scikit-learn, FAISS, YOLOv8, NLTK, pandas, NumPy, Matplotlib

**Tools Git**, GitHub, Azure DevOps, Jira, Selenium

## SELECTED PROJECTS

**Dynamic No-Code Web Platform** - *Laravel, React, Inertia.js*

- Administrative interface allowing non-technical users to create, modify and manage database schema and fields at runtime without code changes.

**Real-Time Chat Application** - *Flask, Socket.IO*

- Multi-format messaging supporting text, images, voice recordings and file sharing over persistent WebSocket connections.

**Fingerprint-Based Attendance System** - *Raspberry Pi, Arduino, Linux*

- Hardware-integrated attendance system with biometric capture and a web interface delivering instant server-side updates.

## CERTIFICATIONS & AWARDS

- Winner - Saegis Campus Hackathon (2025)
- Machine Learning Specialization - DeepLearning.AI, Coursera (2025)
- Django REST Framework - Meta, Coursera (2025)
- ASP.NET - Udemy (2025)
- Laravel: Basic to Advanced - Udemy (2024)
- PostgreSQL: Basic to Advanced - Udemy (2023)

## REFERENCES

*Available on request.*